Security Requirements and Inspection Tool Development for 5G Private Network Deployment in Smart Factories*

Jieon Oh[1] and Ilsun You[12†]

[1] Kookmin University, Seoul, Republic of Korea
[2] KMU Global Research Center for ICT Convergence Security, Kookmin University, Seoul, Republic of Korea
{jieon1006, isyou}@kookmin.ac.kr

Abstract

This paper defines production continuity as a key security requirement for smart factories deploying 5G NPN (Non-Public Network) and develops an automated inspection tool that verifies AMF/SEAF (Access and Mobility Management Function/Security Anchor Function) compliance with the authentication exception-handling procedures specified in 3GPP (3rd Generation Partnership Project) TS 33.501. The tool evaluates whether the core network properly handles authentication failure scenarios that may affect production continuity. Experiments conducted on an Open5GS-based testbed demonstrated that Open5GS correctly implemented the required exception-handling mechanisms and satisfied all inspection items.

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†Corresponding Author